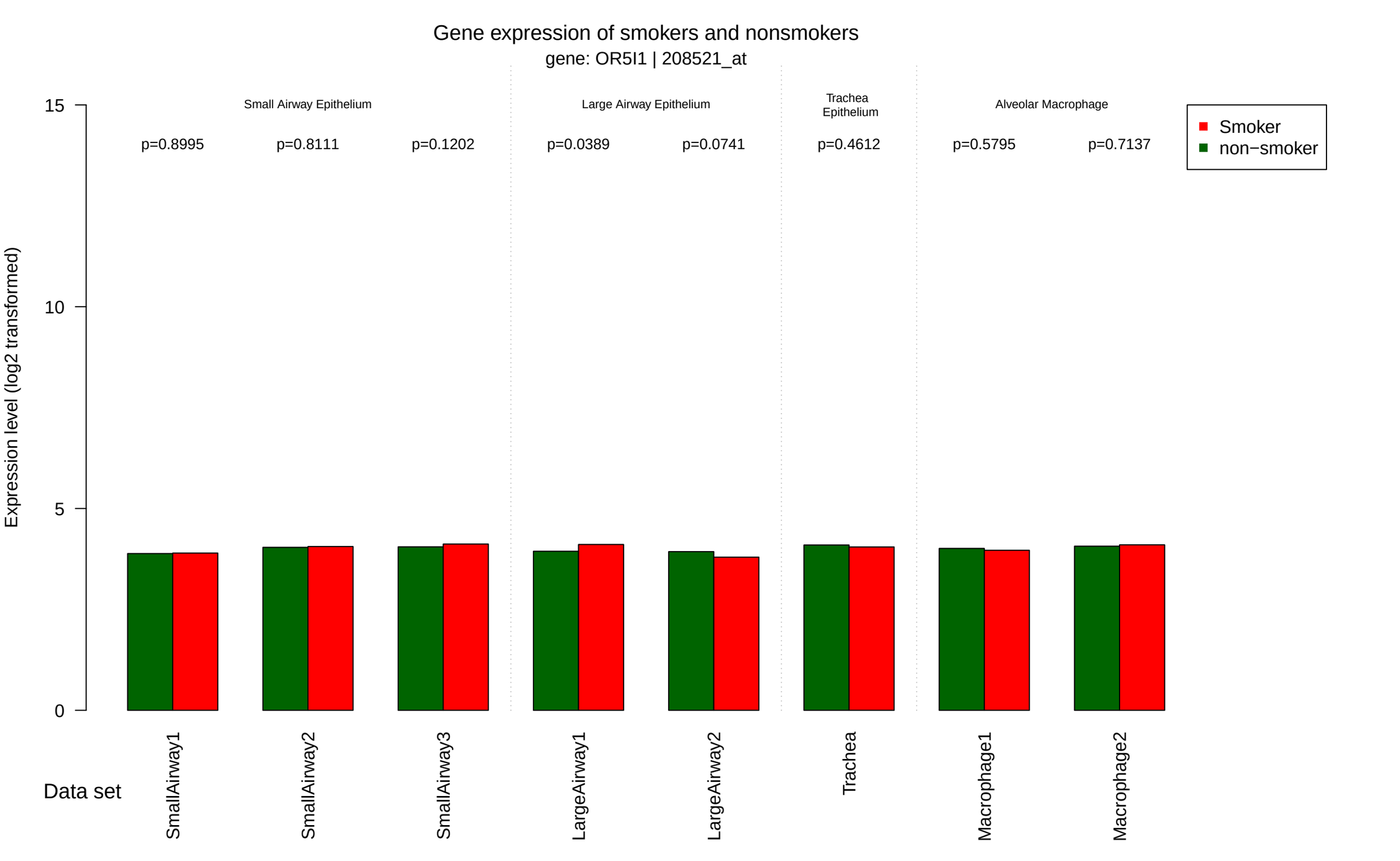
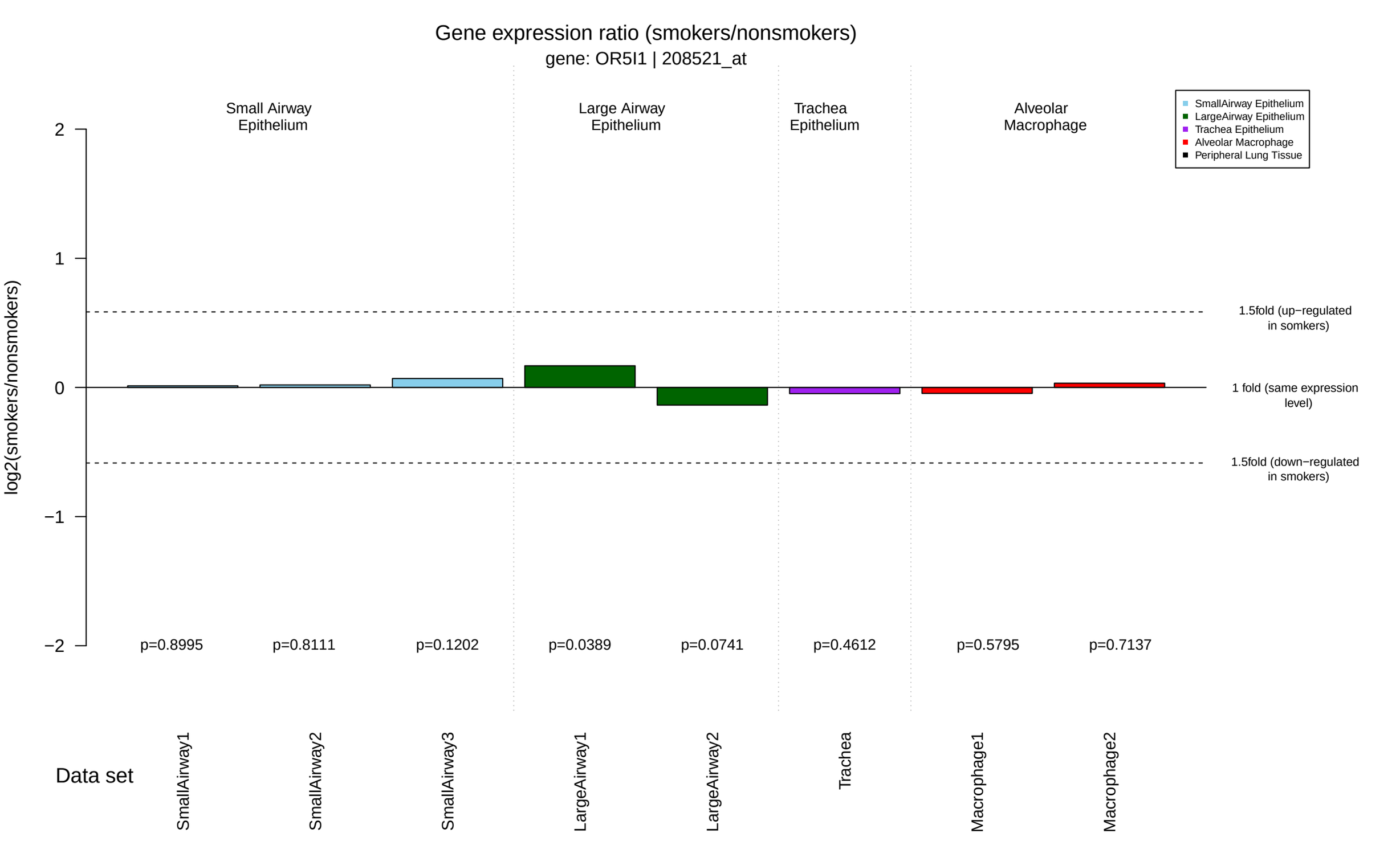
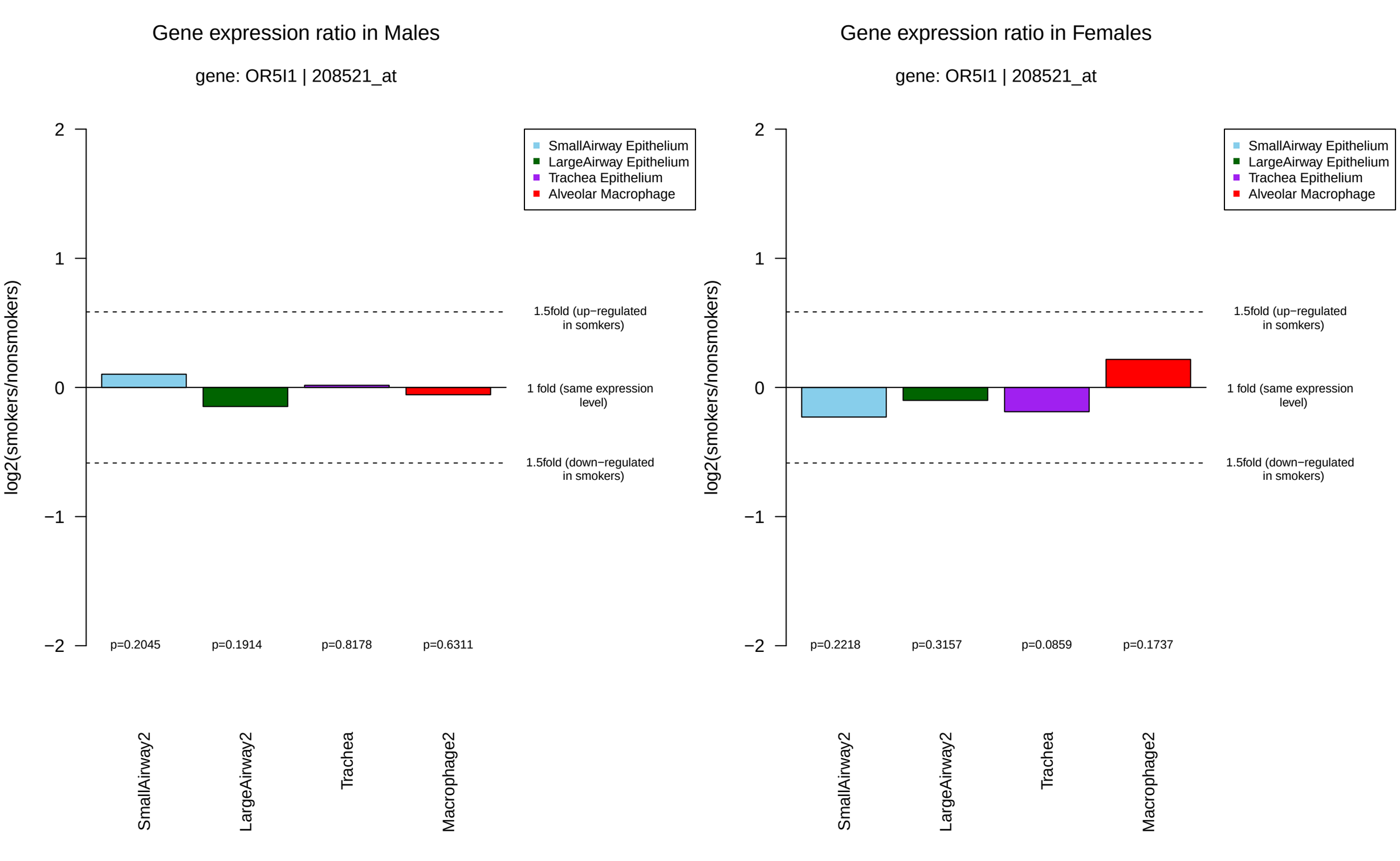
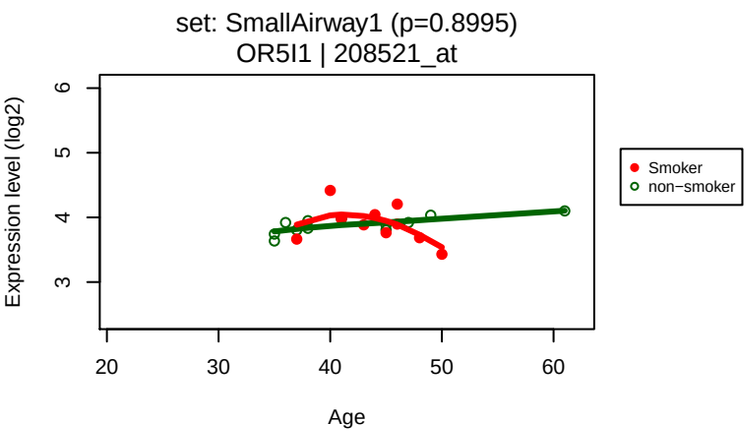
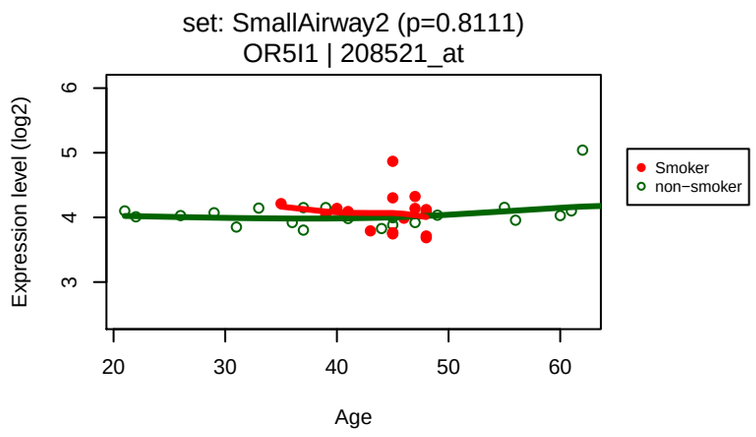
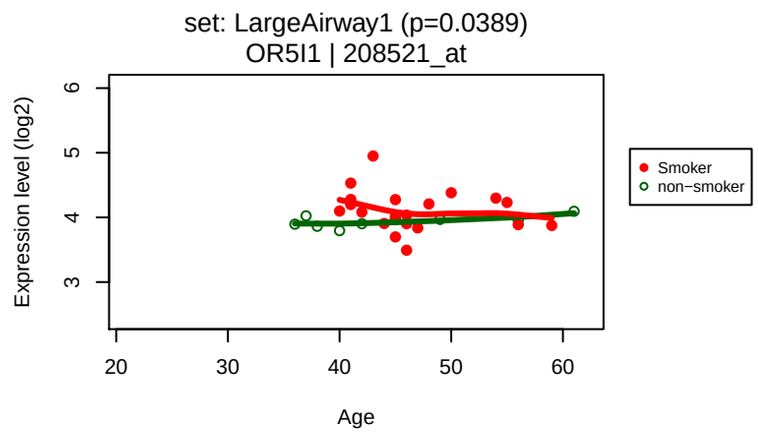

Gene expression of smokers and nonsmokers
gene: OR5I1 | 208521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8995
p=0.8111
p=0.1202
p=0.0389
p=0.0741
p=0.4612
p=0.5795
p=0.7137
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

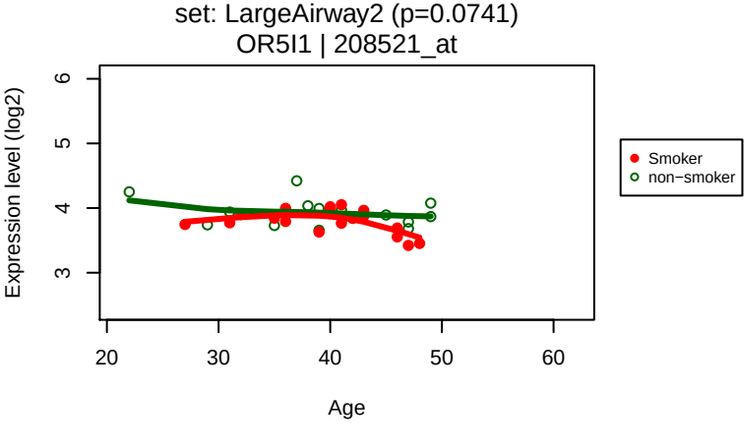

Gene expression ratio (smokers/nonsmokers)
gene: OR5I1 | 208521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8995
p=0.8111
p=0.1202
p=0.0389
p=0.0741
p=0.4612
p=0.5795
p=0.7137
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

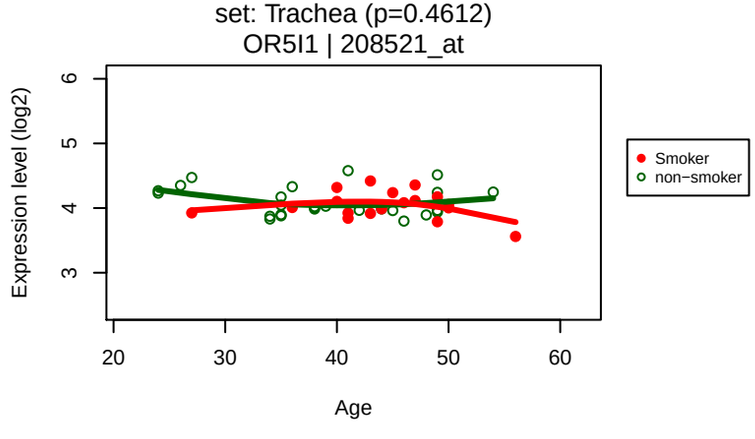

Gene expression ratio in Males
gene: OR5I1 | 208521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2045
p=0.1914
p=0.8178
p=0.6311
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR5I1 | 208521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2218
p=0.3157
p=0.0859
p=0.1737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

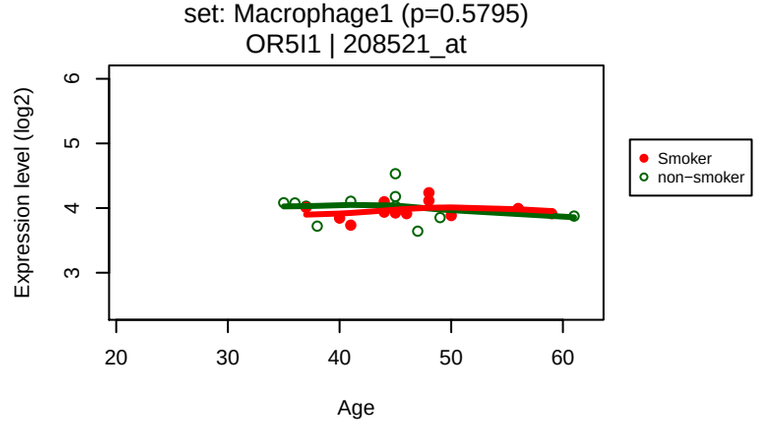

set: SmallAirway1 (p=0.8995)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker

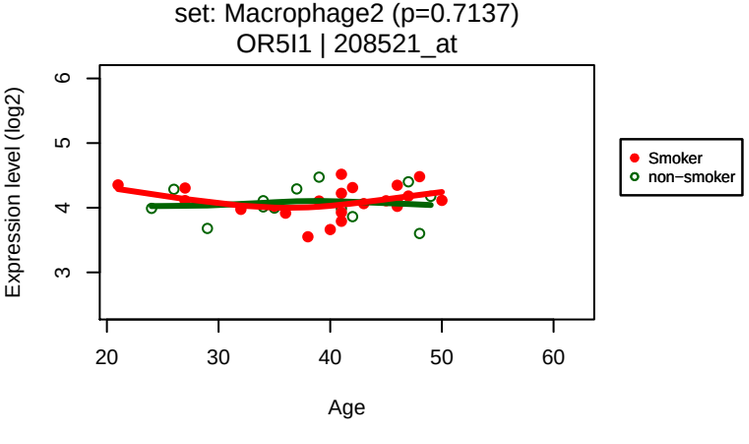

set: SmallAirway2 (p=0.8111)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker

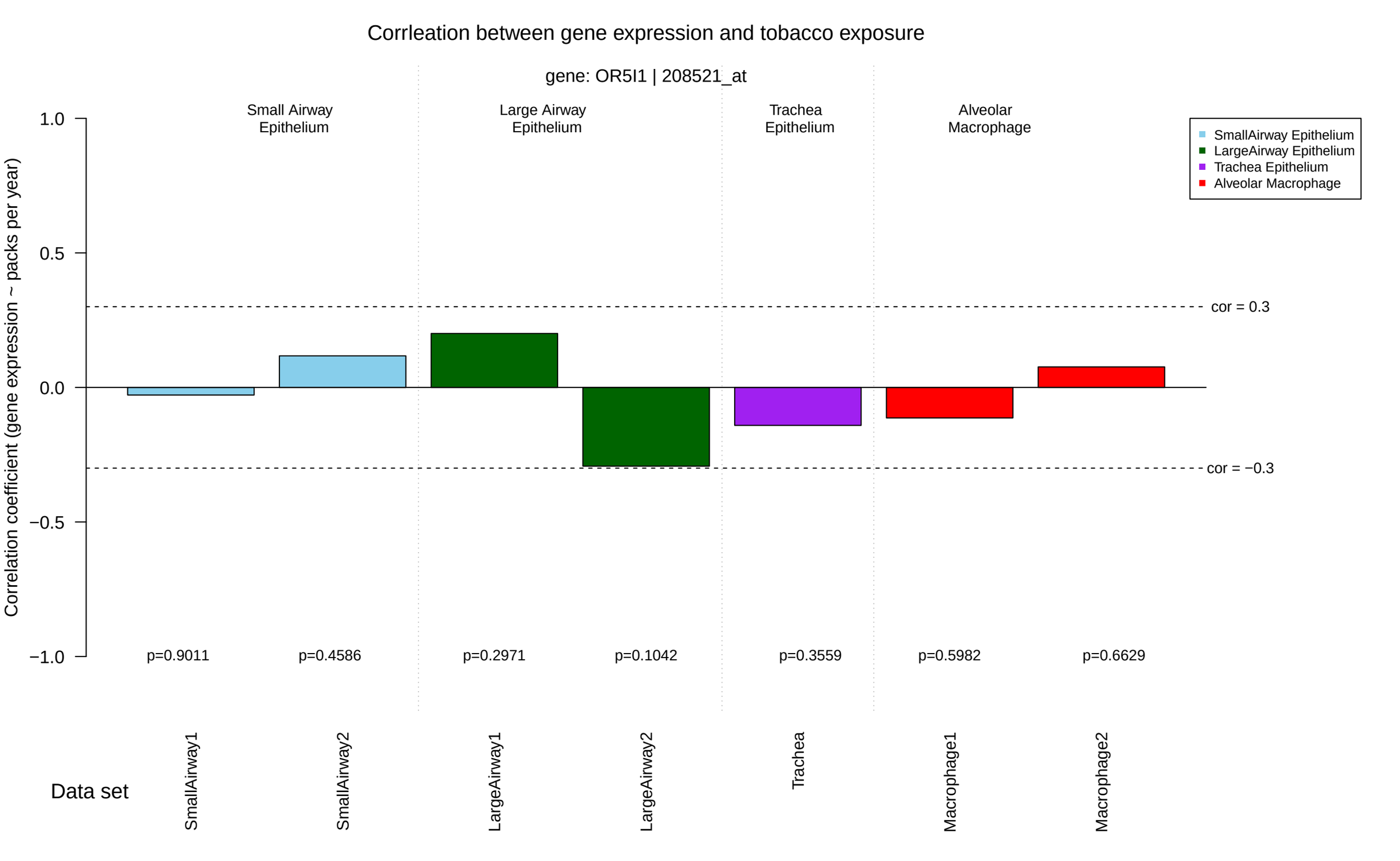

set: LargeAirway1 (p=0.0389)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0741)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4612)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5795)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7137)
OR5I1 | 208521_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR5I1 | 208521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9011
p=0.4586
p=0.2971
p=0.1042
p=0.3559
p=0.5982
p=0.6629
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2